Sri Charan Rachakonda

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AI/ML ENGINEER & DATA SCIENTIST

Data Scientist and AI/ML Engineer with 4 years of enterprise experience building production-grade AI and ML systems across Banking, Healthcare, and Insurance. Specializes in clinical fraud detection, LLM-based document intelligence, and customer analytics with end-to-end ownership from model development through cloud deployment on AWS and Azure. Consistent track record of reducing manual effort, improving model accuracy, and delivering measurable operational outcomes.

TECHNICAL SKILLSET

- **Programming Languages:** Python, SQL
- **Data Analysis & Visualization:** NumPy, Pandas, SciPy, Matplotlib, Seaborn, Power BI, Tableau
- **Machine Learning:** Scikit-learn, XGBoost, Random Forest, Logistic Regression, Gradient Boosting, Clustering, Anomaly Detection
- **Deep Learning & NLP:** PyTorch, TensorFlow, Transformers, BERT, NER, Sentiment Analysis, Information Extraction
- **Generative AI:** LLMs, Retrieval-Augmented Generation (RAG), Prompt Engineering, LangChain, Microsoft AutoGen, LoRA/QLoRA Fine-tuning
- **Vector Databases:** FAISS, Neo4j
- **MLOps & Automation:** MLflow, Airflow, Kubernetes, Docker, FastAPI, Flask, GitHub Actions, n8n, Cursor AI, MCP
- **Cloud:** AWS (S3, SageMaker, Bedrock, Textract, Glue), Azure (Azure ML, AKS, Azure OpenAI)
- **Evaluation & Safety:** SHAP, LIME, RAGAS, BLEU/ROUGE, ROC-AUC, Prompt Safety Testing, Bias Analysis
- **Security & Compliance:** HIPAA, PII/PHI Redaction, HITL Review Loops

PROFESSIONAL EXPERIENCE

Wells Fargo | USA
Gen AI Engineer / Data Scientist **May 2024 - Present**

Project: Intelligent Financial Document Intelligence & Automated Regulatory Reporting Platform

Built an end-to-end GenAI platform for Wells Fargo's corporate banking division to automate extraction, structuring, and narrative generation from 50,000+ financial documents annually, replacing manual analyst workflows with LLM-driven pipelines integrated with Graph-RAG retrieval and agentic orchestration.

- Engineered an enterprise document intelligence pipeline using AWS Textract to extract and structure data from 50,000+ unstructured financial reports annually, reducing manual data preparation effort by 40% and improving downstream analytics consistency.
- Built OCR extraction workflow with field-level validation and audit trails, ensuring consistently compliant, audit-ready inputs for regulatory reporting across 12+ financial document types.
- Implemented Graph-RAG architecture using Neo4j as the knowledge store, increasing factual accuracy of LLM-generated financial reports by 35% over standard dense retrieval baselines.
- Developed LLM-based automated report generation using LangChain, cutting recurring analyst report turnaround from 2 days to under 4 hours and eliminating manual narrative authoring for 8+ report types.
- Architected multi-step agentic workflows using Microsoft AutoGen to orchestrate data retrieval, context assembly, and report generation, with human-in-the-loop validation checkpoints enforced for all regulated outputs.
- Applied LoRA/QLoRA fine-tuning to adapt foundation models for Wells Fargo's financial vocabulary, reducing domain-specific terminology errors by 28% and improving regulatory citation precision on held-out evaluation sets.
- Deployed GenAI and ML workloads on AWS SageMaker and Bedrock with autoscaling inference endpoints, sustaining sub-5-second response latency across 500+ concurrent analyst queries at peak volumes.
- Defined model evaluation criteria covering accuracy, factual consistency, and latency SLAs in alignment with compliance and finance stakeholders, reducing post-deployment rework cycles by 60%.

Max Healthcare | India
Data Scientist (ML) **July 2022 – July 2023**

Project: Healthcare Claims Fraud Detection & Clinical NLP Analytics System

Developed and deployed a production ML fraud detection system for Max Healthcare's medical claims operations, combining ensemble models, LSTM temporal pattern analysis, and BERT-based clinical NLP to identify fraudulent billing across a network of 200+ hospitals at AUC-ROC 0.88.

- Designed and deployed a healthcare fraud detection system using XGBoost, Isolation Forest, and one-class SVM on Max Healthcare's medical claims data, achieving an AUC-ROC of 0.88 on held-out validation sets.
- Engineered a 120+ variable feature matrix from ICD-10 diagnosis codes, CPT procedure codes, provider identifiers, and billing patterns, improving model signal quality and lifting AUC by 0.06 over baseline features.
- Built a BERT-based NLP pipeline to mine clinical notes and provider narratives for anomalous billing descriptions, surfacing text-derived fraud signals that improved precision by 30% over keyword-baseline methods.
- Applied an LSTM-based temporal sequence model to capture longitudinal billing patterns across provider-patient histories, improving fraud detection recall by 18% over static feature-only baselines.
- Resolved class imbalance in fraud labels (4% positive rate) using SMOTE and cost-sensitive learning, improving minority-class recall by 22% over uncalibrated baselines.
- Deployed all ML models to production on Azure ML and AKS with real-time fraud scoring via Flask APIs, integrated directly into Max Healthcare's clinical management systems serving 200+ hospitals.
- Delivered SHAP and LIME interpretability reports to clinical compliance stakeholders, enabling explainable fraud decisions that were accepted by the internal audit team with minimal follow-up.

SBI Life Insurance | India
Data Scientist **July 2021 – May 2022**

Project: Customer Churn Prediction, Policyholder Segmentation & Targeted Marketing Analytics Platform

Built an end-to-end churn prediction and customer segmentation platform for SBI Life Insurance's retail business unit, combining gradient-boosted classification models and k-means clustering on 3M+ policyholders to power data-driven targeted marketing campaigns with measurable conversion uplift.

- Developed Random Forest and Gradient Boosting models to forecast customer churn and purchase likelihood, achieving 82% accuracy and 19% improvement in churn identification over the prior rules-based system.
- Segmented the policyholder base into 6 behavioural cohorts using k-means clustering, enabling targeted campaigns with a measured 23% uplift in conversion versus broadcast campaigns.
- Engineered 80+ features from policy tenure, premium payment patterns, and product interaction history, lifting model F1 score by 11% and reducing false-negative churn misclassifications by 16%.
- Built and optimized AWS Glue ETL pipelines processing 10M+ policy and claims records monthly, reducing data processing time by 30% and enabling weekly model refresh cycles.
- Applied SageMaker Bayesian hyperparameter optimization across model iterations, achieving a 15% improvement in prediction accuracy versus default configurations across all policyholder segments.
- Designed A/B tests comparing ML-driven targeting against rule-based campaigns, demonstrating a 17% improvement in campaign ROI and statistically significant retention uplift across 4 policyholder segments.
- Built Power BI dashboards visualising churn risk scores, segment distributions, and campaign performance, used by marketing and product leadership for quarterly planning across 3M+ policyholders.

EDUCATION

- **Masters in Advanced Data Analytics** | University of North Texas
- **Bachelors in Computer Science and Engineering** | Bharath Institute of Higher Education and Research

CERTIFICATIONS

- **Microsoft Certified**: Azure AI Engineer Associate
- **Oracle**: Generative AI Professional